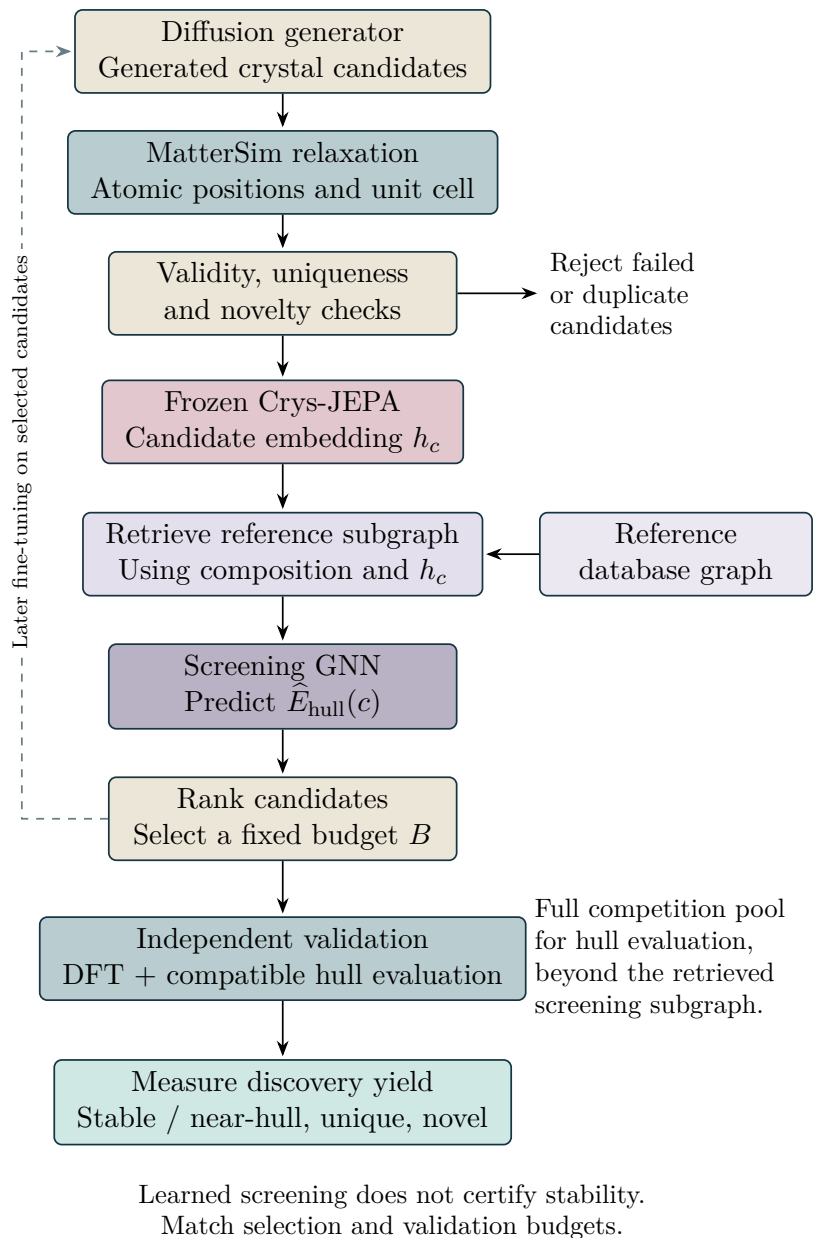
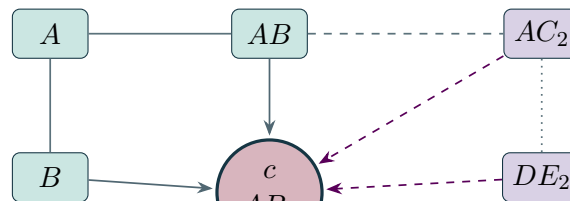


(a) Candidate screening and validation



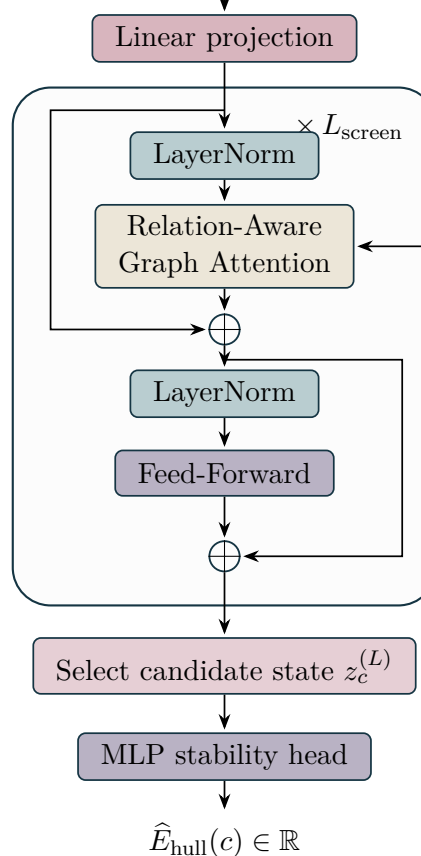
(b) Candidate-conditioned screening GNN

Relaxed candidate c and retrieved whole-crystal references



Pink: candidate query node
Teal: competition references
Purple: cross-chemistry analogues
Solid links: chemical context
Dashed links: analogue context
Dotted links: structural similarity

For candidate and reference nodes:
 $x_i = [h_i^{\text{Crys-JEPA}}; \text{composition}_i]$



Proposed flow. Energy-above-hull regression is illustrated; the objective and graph-attention cell remain design choices.